

# Epigenetics

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# Epigenetic

Epigenetic phenomena: heritable alternative states of gene activity that do not result from altered nucleotide sequence



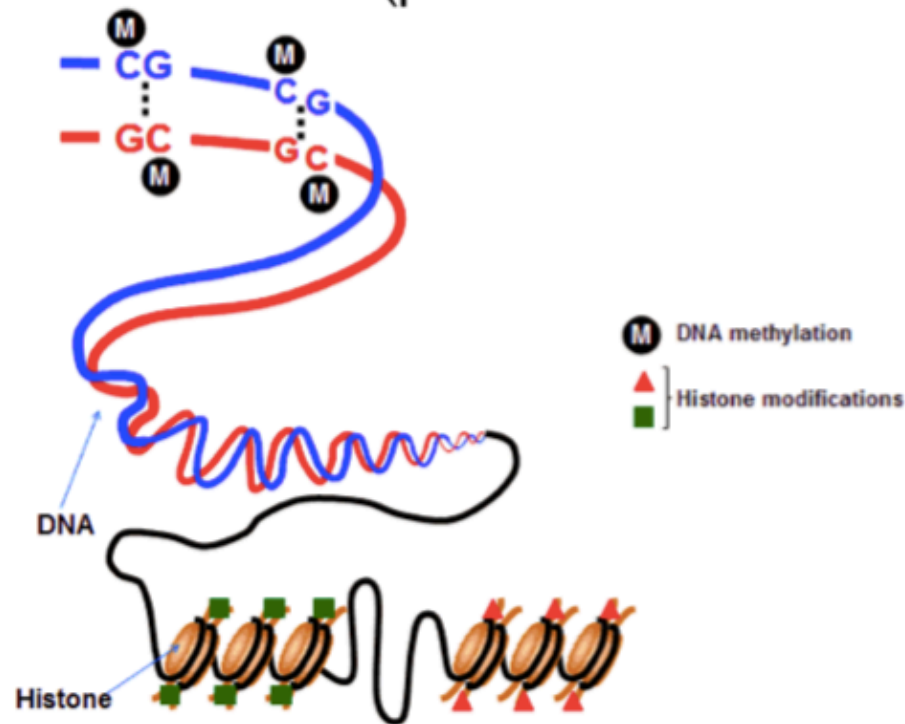
# Definitions

- “Epi” – over, above, outer
- Epigenetics – stably heritable phenotype changes in a chromosome without alterations in the DNA sequence
  - Histone modifications
  - DNA methylation
- Epigenomics – refers to the study of the complete set of epigenetic alterations
- “Epigenetic code” – epigenetic features that maintain different phenotypes in different cells

# Epigenetics

Modification to DNA  
– DNA methylation

Modification to histones  
(proteins around which DNA is wound)



These modifications change

- during differentiation
- as a response to environment

# How Does Epigenetics Work?

- **DNA Methylation**

DNA methylation works by adding a methyl group to DNA.

Typically, this group is added to specific places on the DNA, where it blocks the proteins that attach to DNA to “read” the gene.

This methyl group can be removed through a process called demethylation.

Typically, methylation turns genes “off” and demethylation turns genes “on.”

- **Histone modification**

DNA wraps around proteins called histones.

DNA wrapped tightly around histones cannot be accessed by proteins that “read” the gene. Some genes are wrapped around histones and are turned “off” while some genes are not wrapped around histones and are turned “on.”

Chemical groups can be added or removed from histones and change whether a gene is unwrapped or wrapped (“on” or “off”).

# How Does Epigenetics Work?

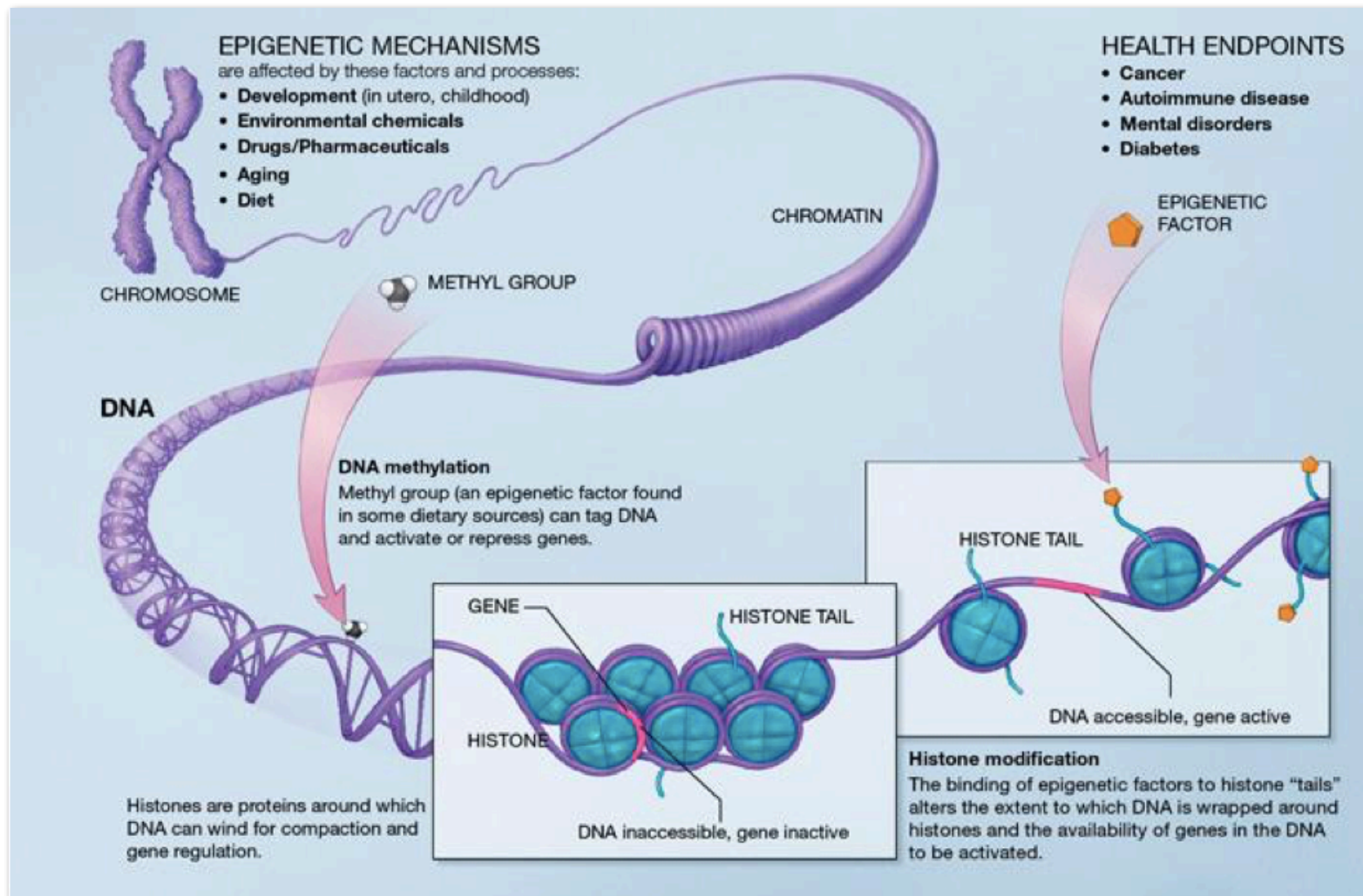
- **Non-coding RNA**

Our DNA is used as instructions for making coding and non-coding RNA.

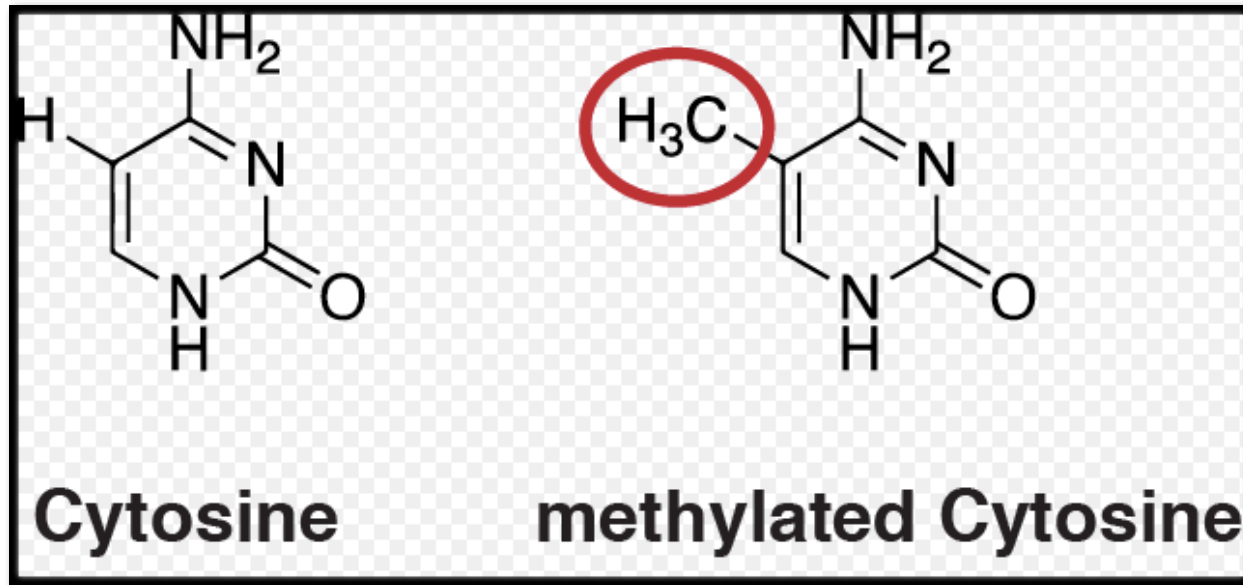
Non-coding RNA helps control gene expression by attaching to coding RNA, along with certain proteins, to break down the coding RNA so that it cannot be used to make proteins.

Non-coding RNA may also engage proteins to modify histones to turn genes “on” or “off.”

# EPIGENETICS OVERVIEW



## METHYLATION ALWAYS FOR CYTOSINE

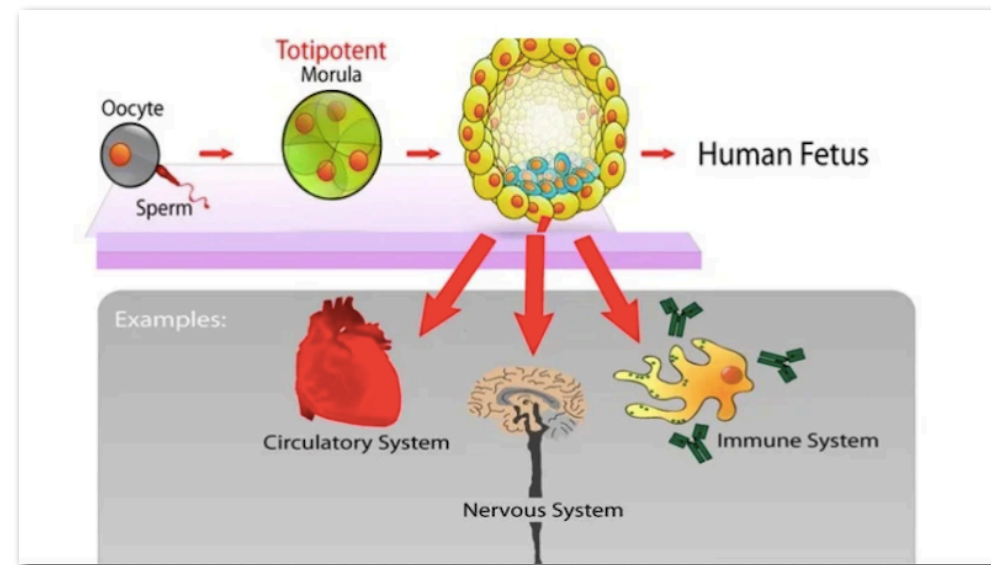




# How Can our Epigenetics Change?

## 1. Epigenetics and Development

Epigenetic changes begin before we are born. All our cells have the same genes but look and act differently. As you grow and develop, epigenetics helps determine which function a cell will have, for example, whether it will become a heart cell, nerve cell, or skin cell.

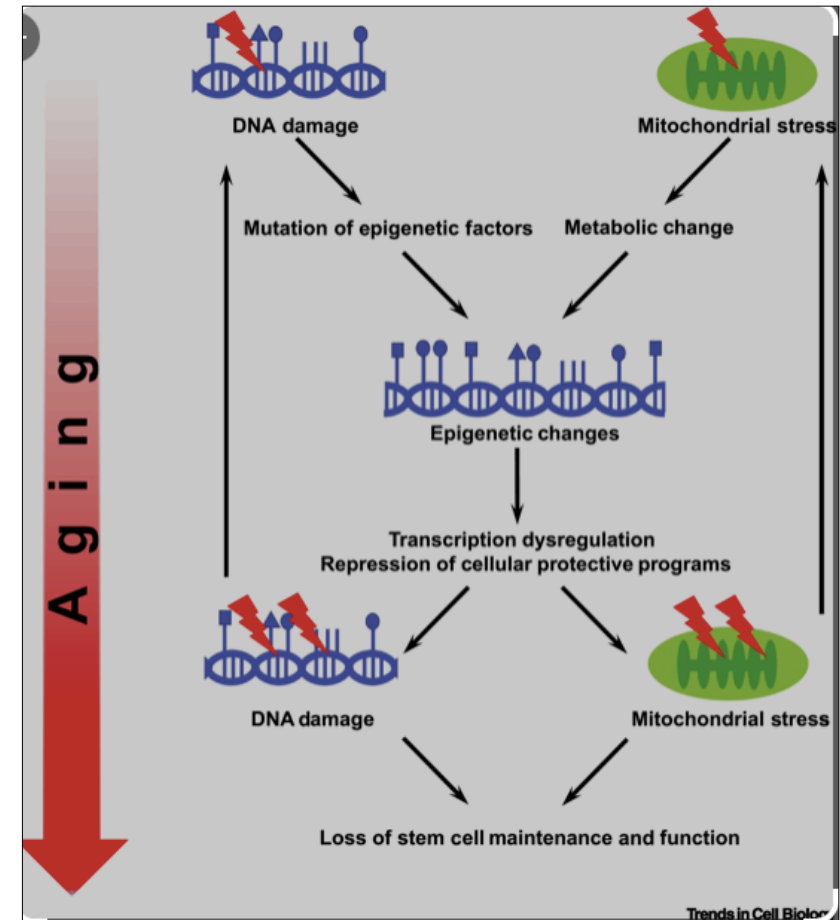


# How Can our Epigenetics Change?

## 2. Epigenetics and Age

our epigenetics change throughout the life.

The epigenetics at birth is not the same as epigenetics during childhood or adulthood.



Trends in cell biology

# Epigenetics and Reversibility

Not all epigenetic changes are permanent. Some epigenetic changes can be added or removed in response to changes in behavior or environment.

Example: heavy smokers vs. Non-smokers vs. Former smokers

Smoking can result in epigenetic changes. For example, at certain parts of the *AHRR* gene, smokers tend to have less DNA methylation than non-smokers.

**AHRR** is known to inhibit the aryl hydrocarbon receptor, which is important to metabolizing harmful chemicals.

The resultant increase in **AHRR** expression could lead to a decrease in the body's ability to break down carcinogens, increasing the risk of cancer.

# Why is Methylation Important?

Proper methylation is essential for life and is involved in many functions in the body such as:

- Gene expression
- Fetal development
- Production of neurotransmitters
- Maintaining proper homocysteine levels
- Detoxification and processing of heavy metals like arsenic
- Production of glutathione, the body's master antioxidant
- Production of crucial compounds like [creatine](#), [phosphatidylcholine](#) and [carnitine](#).
- Maintaining proper histamine levels
- Mental and brain health
- Nervous system health
- Cardiovascular health
- Mitochondrial health
- Reproductive health
- Liver health

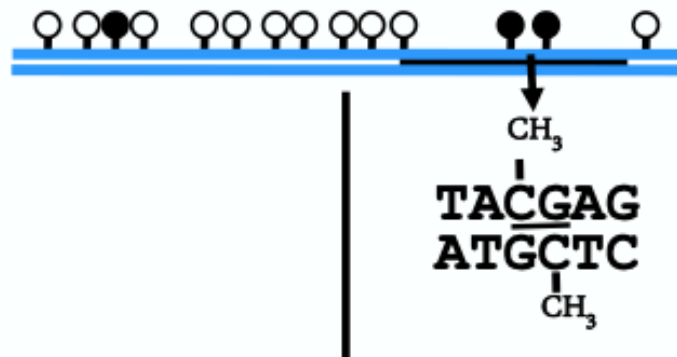
# How to Optimize Your Methylation

We should be doing everything we can to optimize our body's ability to effectively methylate because it's such a critical process. It's not that one can simply do it once and be done. Methylation must be continually maintained throughout life,. Problems start arising when methylation is not optimized, meaning, it's happening slower than it should be (or the opposite: it's happening too fast!). In general, the methylation cycle slows down when it doesn't have enough methyl-supportive nutrients and speeds up when it has too many.

## **Important methylation support nutrients include:**

- Riboflavin
- Vitamin B6
- Methylfolate
- Vitamin B12 in the form of Methylcobalamin
- Choline
- Betaine (trimethylglycine, TMG)
- Magnesium
- Zinc

## DNA Modification



### DNA methylation

Alterations



Cell division



Progeny